ANH NGUYEN HOANG

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RESEARCH INTERESTS

My current research focuses on AI for Software Engineering, particularly foundation models for code, autonomous coding agents, and long-horizon software evolution. I aim to study principled approaches for training and adapting code-focused language models, as well as designing reliable agentic systems for repository-scale software engineering tasks. I am also open to diversifying my research across related areas in the future.

EDUCATION

Hanoi University of Science and Technology 2020 – 2024
Bachelor of Science in Computer Science
GPA: 3.75/4.0 Ranked 15th of 296 students (Top 5%)

PUBLICATIONS

[1] **CodeWiki: Evaluating AI's Ability to Generate Holistic Documentation for Large-Scale Codebases**
Anh Nguyen Hoang, *Minh Le-Anh, Bach Le, Nghi D. Q. Bui*
Findings of ACL 2026 [Paper] [Code]

[2] **Beyond Repository Boundaries: Cross-Repository Graph Retrieval for Code Generation**
Minh Le-Anh, Nam Le Hai, Quyen Tran, ***Anh Nguyen Hoang****, Linh Ngo Van, Bach Le, Nghi D. Q. Bui*
Findings of EMNLP 2026

[3] **Mutual-Pairing Data Augmentation for Few-Shot Continual Relation Extraction**
Nguyen Hoang Anh*, Quyen Tran*, Thanh Xuan Nguyen*, Nguyen Thi Ngoc Diep, Linh Ngo Van, Thien Huu Nguyen, Trung Le*
NAACL 2025 [Paper] [Code] ** denotes equal contribution.*

[4] **Preserving Generalization of Language Models in Few-Shot Continual Relation Extraction**
Quyen Tran, Nguyen Xuan Thanh*,* ***Nguyen Hoang Anh****, Nam Le Hai, Trung Le, Linh Van Ngo, Thien Huu Nguyen*
EMNLP 2024 [Paper] [Code]

PREPRINTS AND UNDER REVIEW

[1] **FALCON: A Counterexample-Grounded Verification Framework for Secure Code Generation**
Tue Le, ***Anh Nguyen Hoang****, Gabriel Sasseville, Doseok Jang, Pierre-Luc Bacon, Nghi D. Q. Bui*
Under review

[2] **SecPRBench: Can Language Model Agents Detect Vulnerabilities in Code Changes?**
Yihan Yang, Chenyuan Yang, Zijie Zhao, ***Anh Nguyen Hoang****, Ziqi Zhang, Weidong Wang, Zhiheng Lyu, Lingming Zhang*
Under review

[3] **RustPrint: Documentation-Guided Agentic Codebase Migration from C to Rust**
Minh Le-Anh, ***Anh Nguyen Hoang****, Bach Le, Nghi D. Q. Bui*
Preprint, 2026 [Preprint]

HONORS AND AWARDS

Champion, AI for Real Action 2025 January 2026
First prize with Team Converged among nearly 400 teams in Vietnam's first national artificial intelligence competition

- **Recipient, Techcombank Overseas PhD Study Sponsorship** April 2026
  Sponsorship limit of USD 333,000 for overseas PhD study [Offer Letter]

EXPERIENCE

FPT Software AI Center May 2025 – Present
AI Research Resident *Hanoi, Vietnam*
Advisors: Associate Prof. Bach Le, University of Melbourne; Assistant Prof. Nghi D. Q. Bui, VinUniversity

- Developed CodeWiki, an LLM-based framework that generates holistic, repository-level documentation for large software projects
- Co-developed FALCON, a counterexample-grounded verification framework that provides execution-grounded security feedback for secure code generation, used for test-time repair and as an RL reward
- Contributed to SecPRBench, a benchmark of 301 security review tasks from disclosed vulnerabilities for evaluating language model agents on detecting vulnerabilities in real code changes
- Research documentation-guided code migration and automated program repair

Research Assistant March 2024 – April 2025
Continual Learning and Natural Language Processing Research *Hanoi, Vietnam*
Advisors: Assistant Prof. Linh Ngo Van; Associate Prof. Thien Huu Nguyen

- Researched few-shot continual relation extraction and generalization, resulting in equal-contribution papers at EMNLP 2024 and NAACL 2025

Oraichain Labs May 2024 – April 2025
AI Engineer *Hanoi, Vietnam*

- Led the development of Cupiee, an emotional AI companion platform with more than 300,000 users and approximately 100,000 messages per day
- Designed autonomous multi-agent systems for social media engagement and market analysis

SELECTED OPEN-SOURCE SOFTWARE

CodeWiki 1,700+ GitHub stars
Open-source repository documentation framework supporting seven programming languages

XTTSv2-Finetuning-for-New-Languages 200+ GitHub stars
Toolkit for adapting the XTTSv2 speech-synthesis model to new languages

LANGUAGES

Vietnamese Native proficiency
English Professional working proficiency (IELTS 7.0)

REFERENCES

Associate Prof. Bach Le
University of Melbourne, Australia
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Assistant Prof. Nghi D. Q. Bui
Staff Research Scientist, Google Research
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Associate Prof. Thien Huu Nguyen
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